OSCAR DE LA CERDA

Portfolio: oadlc.github.io
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EDUCATION

Yale University, New Haven, CT — May 2028

- *ABET-accredited B.S in Electrical Engineering, Certificate in Programming | GPA: 3.8*
- *Relevant Coursework:* Electromagnetism, Introduction to Computer Engineering, Introduction to Communications and Control, Circuits and Systems Design, Data Structures and Programming, Digital Systems

University of Iceland, Reykjavik, Iceland — June – July 2025

- *Relevant Coursework:* Renewable Energy Technology, Politics and Economics of Renewables

TECHNICAL SKILLS

- *Electromechanical Hardware Design: Schematic Capture (Altium, KiCad), custom circuit design, PCB layout & assembly, analog/digital circuit design, ADC circuits, instrumentation & controls, sensor integration (GPS, IMU, altimeter), power systems, FPGA/HDL design*
- *Controls & Simulation: Closed-loop / PID control system design, control-loop tunning and debugging, Zemax OpticStudio, LTspice, MATLAB*
- *Languages & Tools: Python, MATLAB, C, C++, Verilog, Racket, PowerShell; Altium, KiCad, LTspice, Zemax; building Siemens NX proficiency through L'SPACE Mission Concept Academy*

EXPERIENCE

Yale Undergraduate Aerospace Association, *Avionics Sub team member* — November 2024 - Now

- Designed, simulated, and implemented an analog-to-digital converter (ADC) circuit to enable a Teensy microcontroller to process pressure, voltage, and current sensor data during pre-fuel testing, directly supporting instrumentation & controls for flight readiness
- Developed a complete avionics sensor-suite schematic integrating GPS, IMU, altimeter, MCU, and S-PLIG modules to improve radio/communications and circuitry reliability and flight data acquisition

Quantum Network Research, *Undergraduate Researcher* — June 2026 - Now

- Automated fast-steering-mirror (FSM) sweep sequences across a quad-cell optical/radiation-detection sensor in Python, mapping boundary conditions and characterizing sensor response across the full field of view
- Designed and implemented a closed-loop control system to stabilize laser pointing on a quad-cell sensor under simulated FSM disturbances, applying characterized sensor boundaries to inform controller design

Yale Bulldogs Racing, *Sub team Member* — November 2024 – May 2025

- Assembled and soldered a high-voltage PCB for and its circuitry components for the team's Formula-style electric race car, supporting power-delivery reliability in competitive environments

PROJECTS

Two-Axis Ball-Balancing Stabilizer Platform, *Independent Project* — June – July 2026

- Designed a 3RPS-style two axis stabilizer around a custom ESP32-WROVER-CAM PCB carrier board (schematic capture and layout in KiCad) driving three MG966R servos (control equipment) – end-to-end hardware, firmware, and controls integration
- Debugged firmware level GPIO/PSRAM and PWM channel conflicts during board bring-up, and migrated servo control from HTTP to UDP to eliminate control-loop latency
- Built a Python/OpenCV vision pipeline with HSV ball tracking feeding a dual-axis PID controller driven by a law of cosines inverse-kinematics solver